Md. Minhazul Haq

Graduate Research Assistant, Universiti Malaya, Malaysia | Portfolio | Google Scholar | GitHub
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Research Interests

Embedded AI, Smart Transportation Systems, Computer Vision, AI for Cyber-Physical Systems, LLMs

Publications

1. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi, Hafiz Muhammad Fahad Noman, and Mohamad Haniff Junos. *HAT-YOLO: Hybrid Attention and Transformer-Integrated Model for UAV Object Detection on Embedded Devices.* KSII Transactions on Internet and Information Systems, 20(2):920–941, 2026.
2. Yang Z, Khairuddin AS, Huang CJ, Ru WW, Noman HM, Putri TW, Xu X, Haq MM. *A Review of Short-Term, Long-Term, and Memory Consolidation Mechanisms in the Hippocampus and Cerebral Cortex.* IEEE Access, 2025.
3. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. *EdgeLite-YOLOv8n: An Efficient YOLOv8n Model for Real-Time Vehicle Detection in UAV Imagery on Embedded Platforms.* Accepted at the 2nd International Transdisciplinary Research Conference 2026, Universiti Malaya.
4. Md Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. An Optimized YOLOv8n Model for Real-Time Small-Object Detection on Resource-Constrained Platforms. Review in progress (Multimedia Tools And Applications Journal), Submission Id:- MTAP-D-26-02236.
5. Md Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. Towards Real-Time Wood Defect Detection on Resource-Constrained Devices: An Optimized YOLOv8n Model. Review in progress (Forest Science), Submission Id:- FRSC-D-26-00118.
6. Hasan, M. M., K, S., A, M., & Haq, M. M. *Development of Prototype Dual-Axis Solar Tracking Device for Optimal Power Generation at Residential Area in Bangladesh.* Journal of Energy and Safety Technology, 7(1):25–33, 2024.

Poster Presentation: Md. Omar Faruk, Md. Minhazul Haq, Anis Salwa B. M. Khairuddin, Effariza Hanafi, Karen L. Pua, and W. M. Tilakaratne. *Real-Time Oral Cancer Detection Using Lightweight YOLOv8n-Based Deep Learning and Computer Vision: A Pilot Study.* UMHEALTH Academic Month & Research Carnival 2025, Universiti Malaya.

Research Experience

Graduate Research Assistant Nov. 2023 – Feb. 2026
Centre of Intelligent Systems for Emerging Technology (CISET), Universiti Malaya

- Developed lightweight YOLO-based models for UAV object detection on embedded and edge devices.
- Developed and optimized attention-enhanced lightweight computer vision architectures for real-time vehicle and pedestrian detection on a low-cost, resource-constrained Jetson Nano device.
- Conducted benchmarking and comparative evaluation of SOTA object detection models using multiple benchmark datasets.

Education

Master of Engineering Science (Research) Nov. 2023 – Present
Department of Electrical Engineering, Universiti Malaya, Malaysia
Field of Research: Artificial Intelligence
Thesis: *Optimization of Lightweight YOLO Model for Automated Object Detection on Embedded Device*

B.Sc. in Electrical and Electronics Engineering Final Year GPA: 3.55/4.00 | CGPA: 3.11/4.00
Jashore University of Science and Technology (JUST), Bangladesh Graduated: Apr. 2022
Thesis: *Design of Solar Tracking System for Optimal Power Generation at Residential Area in Bangladesh*

Teaching Experience

Graduate Teaching Assistant / Lab Demonstrator Oct. 2025 – Feb. 2026
LAB 2 (KIE1002), Department of Electrical Engineering, Universiti Malaya

Technical Skills

Programming: Python, C/C++, MATLAB, HTML
AI/ML: YOLO, RF-DETR, CNN, RNN, Computer Vision
Tools: OpenCV, PyTorch, Docker, Embedded Systems (e.g., Jetson Nano)

Awards

- **Best Presenter (Postgraduate Oral Presentation) – Winner**, 4th International Orthopaedic Postgraduate Symposium (IOPS4), Universiti Malaya, 29 Oct. 2025.
- **Best Project Award** – Voice-Controlled Humanoid Robot, Science Fair 2019, Jashore University of Science and Technology, Bangladesh.
- **Best Project Award** – Solar Scooter, Student to Startup 2.0 Campus Hunt, JUST, Bangladesh, Oct. 2019.

Standardized Test Scores

IELTS – April 8, 2025: Overall 7.0 (Speaking 7.5, Listening 7.5, Reading 7.0, Writing 6.0)